

The 2008 Housing Bubble: Forecasting Systemic Risk with its Causes and Effects

Alona Dhal

August 2025

Contents

1	Introduction	3
2	Economic Factors Behind the Bubble	4
2.1	Cheap Credit: The Illusion of Affordability	4
2.2	From Stability to Speculation	5
2.3	Geography of the Bubble	6
2.4	Financial Innovation and Market Structure Changes	6
3	Policy-Driven Causes of Market Instability	7
3.1	The Federal Reserve	7
3.2	Deregulation and Shadow banking	9
3.3	Oversight of Derivatives and Securitization	11
4	The Crisis Unfolds (2007-2008)	12
4.1	Early signs	12
4.2	Bank Failures and Liquidity Freeze	13
4.3	Stock Market Collapse and Global Shockwaves	14
4.4	Emergency Policy Response and TARP	14
5	Consequences and Long-term Effects	16
5.1	Housing Market Collapse and Foreclosures	16
5.2	Bank Failures	16
5.3	Recovery Measures and Changes in Legislation	17
6	Forecasting Financial Bubbles with Machine Learning	18
7	Data Sources and Feature Engineering	19
7.1	Model Design and Evaluation	20
7.2	Results and Interpretation	21
7.3	Risks and Limitations of ML Forecasting	23
8	A Discussion of Current Events	24
8.1	Today's indicators	24
8.2	Commercial Real Estate Risks	25
8.3	Tech Valuations and AI-Led Speculation	25
8.4	Credit Cycles, Interest Rates and Liquidity	26
9	Conclusion	26
	References	28

1 Introduction

In a single week in September 2008, Merrill Lynch, Lehman Brothers, AIG, and Washington Mutual all collapsed, triggering the worst financial crisis since the Great Depression and erasing eight trillion dollars in household wealth.

The United States housing crisis of 2008 has shaped the modern global economy in terms of its rise until the mid-2000s to its violent collapse in 2007-2008. However, the true demise began a decade earlier, fueled by the bipartisan convergence of political goals and market optimism. President Bill Clinton and President George W. Bush strongly believed that broader access to mortgages would strengthen families and communities. In the short term, these policies appeared to succeed, as they increased access to credit. As a result, the national level of homeownership reached a record high of 69.2% by 2004, seemingly validating the bipartisan push for greater mortgage accessibility [18]. But after this honeymoon period, the impact entailed the loss of eight trillion in household wealth and the deepest global recession since the 1930s.

Beginning in 1995, President Clinton, with the hopes of increasing home ownership, worked to push affordability and reduce financial regulation throughout the United States, characterized by the Financial Services Modernization Act of 1999, Commodity Futures Modernization Act of 2000, and SEC Net Capital Rule Change of 2004. The three measures elicited lenders to increase their subprime products through riskier credits, urged firms to trade more mortgage-backed securities (MBS), and allowed the credit default swaps (CDS) market to operate with less transactional transparency. These efforts combined with Alan Greenspan's interest rates, the former chair of the Federal Reserve, lowered the cost of credit, widened its distribution, and falsely urged the country to treat the housing market as risk-free [15, 3, 22].

Continuing this vision, President Bush shared his "Ownership Society" goal in 2002, pledging to add 5.5 million families to the already growing rate of home ownership. By enacting the American Down-Payment Act of 2003, he authorized low-income families to purchase homes with little cash up front [15]. Additionally, the Department of Housing and Urban Development (HUD) raised the low- and moderate-income purchase goal for Fannie Mae and Freddie Mac from 50% to 56% by 2008 and added new sub-goals targeting very low-income families and special affordable multifamily housing [19]. These efforts were falsely seen as beneficial, when in reality they framed rising ownership as proof of economic health, creating political cover for looser lending.

The boom, however, existed on shaky foundations made up of minimal borrower equity and the false belief that the prices would never fall. As teaser rates reset, defaults cascaded through mortgage-backed securities and their derivative wrappers, freezing interbank lending, toppling institutions once deemed "too big to fail," and spilling far beyond U.S. borders. This paper examines

how a decade of well-intentioned housing policies created precursors for financial catastrophe, analyzes the mechanisms through which the crisis unfolded and evaluates the long-term consequences and regulatory responses that shape the financial landscape today.

2 Economic Factors Behind the Bubble

2.1 Cheap Credit: The Illusion of Affordability

A key enabler for this high-yield and low-risk mindset was the sharp decline in long-term mortgage interest rates. According to data from Freddie Mac, the average 30-year fixed mortgage rate fell from around 8% in the mid-1990s to below 6% by 2003, reaching lows of 5.23% by mid-2003 [20]. This drop in the borrowing rate caused the market to seem more affordable, pushing both consumers and investors to purchase properties.

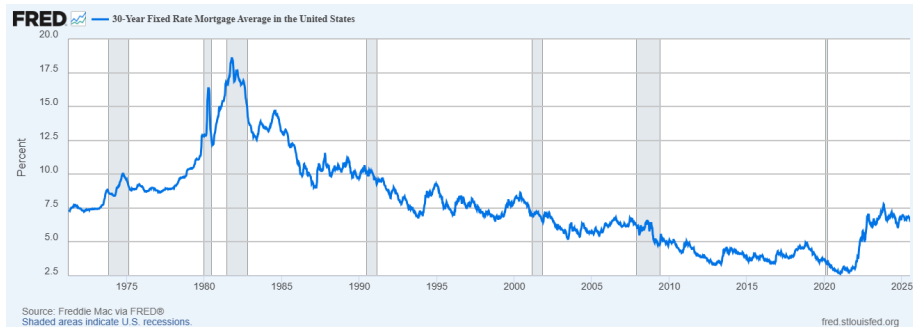


Figure 1: 30-year Fixed Rate Mortgage Average in the United States

However, instead of stabilizing the market, the surge in purchases also caused a direct increase in home prices. The S&P CoreLogic Case-Shiller U.S. National Home Price Index, tracked by the Federal Reserve, shows that from January 2000 to July 2006, home prices rose by over 80%, far outpacing wage growth during the same period [9]. This unsustainable acceleration in prices was especially pronounced in regions like California, Nevada, Arizona, and Florida, where speculative buying was concentrated.

These price surges were not backed by solid equity. According to the Federal Reserve's Flow of Funds Accounts, the median household income remained relatively stagnant throughout the early 2000s, yet conversely, mortgage debt ballooned [1]. Buyers increasingly relied on loan products as adjustable-rate mortgages (ARMs), interest-only loans, and subprime mortgages with minimal documentation [15]. The Consumer Financial Protection Bureau (CFPB) later reported that during the run-up to the crisis, many borrowers were qualifying

for loans with little income verification, pushing the loan-to-value (LTV) ratios to dangerous levels [21]. Meanwhile, the market’s belief in the infallibility of housing prices created moral hazard. Mortgage-backed securities (MBS) were aggressively packaged and sold to investors under the assumption that default risk was negligible because home values would continue to rise [15]. This belief, unchallenged by regulators or rating agencies, created a false sense of security that further fueled speculative activity.

To further build this illusion of affordability, the Federal Reserve aggressively lowered the federal funds rate in the early 2000s, reducing it from above 6% in 2000 to nearly 1% by 2003 [22]. Lower borrowing costs encouraged consumers and investors to purchase more expensive homes, increasing housing demand and driving home prices upward faster than wages. People could now afford to purchase more expensive homes with the same monthly payment, bringing more buyers into the housing market. The increased American purchasing power also drove up home prices faster than wages. From an investor’s perspective, shifting money from low-yielding bonds into real estate and house-flipping seemed appealing, given the low borrowing cost. However, when the rates eventually had to rise and when the teaser rates of the loan products reset, borrowers who could only afford homes at the emergency-low rates were trapped.

While the Federal Reserve’s interest rate cuts after 2001 clearly lowered borrowing costs, former Fed Chair Ben Bernanke pointed to a broader structural force behind the unusually low long-term mortgage rates: a ”global saving glut,” [23].

2.2 From Stability to Speculation

While government policies and deregulation laid the foundation for the housing bubble of 2008, speculative behavior from both consumers and investors spurred its growth. The belief that the prices of real estate properties would rise despite speculative investment and unchecked financial engineering transformed real estate into a speculative asset class. It changed from being a form of stability and personal investment to a vehicle for short-term profit. For the first time, large numbers of borrowers were able to secure mortgages with little to no documentation, minimal down payments, and low initial interest rates. As these behaviors were supported by legislation and policy changes to promote homeownership, they became a standard. Lenders competed to extend credit to riskier borrowers, under the assumption that its effect would be minuscule in comparison to the rising home prices.

Loan products as “No Job, No Income, No Assets” (NINJA), Interest-Only, and Adjusted-Rate Mortgages (ARMs) became part of the standard process. Borrowers could get mortgages without proving they had a stable income, employment, or assets. Alternatively, they could strictly pay the interest for the first few years, with no principal. However, after the interest-only period ends,

the payments would jump exponentially, making it difficult for clients to pay the full amount back [15, 21]. Another option could be to adopt a scheme where the interest rate starts low but adjusts upward over time. Here, the "teaser rates" would start around low and would reset to higher rates after a few years. Even though borrowers planned to refinance or sell before the rates reset, they were trapped once the prices suddenly fell. Similar to the previous type of loan product, the ARMs created a wave of defaults when payments reset higher in 2005-2007 [15, 21].

2.3 Geography of the Bubble

One of the clearest indicators of speculative activity was the rise in non-owner-occupied homes. Investors were looking to purchase these homes solely to make minimal improvements and quickly resell them at a higher price, primarily driven by this artificially inflated demand of the housing market. This activity specifically surged in areas like Phoenix, Miami, and Las Vegas, where the population was known to grow significantly due to new job opportunities, climate, and low cost of living. People looked to purchase and build homes in these cities more than ever, given the tech expansion, job creation, and lower cost of living at the time. Following conventional microeconomic principles, the high demand for houses quickly drove the prices for these homes up. To accommodate the heightened amount of population growth and housing demand, the government put lenient zoning laws into effect, making it easier to buy land, build quickly, and sell fast [17].

To capitalize on the inflated demand, investors bought thousands of homes, but never intended to live in them. The cycle of purchasing a house, making minimal improvements, and then using appreciation to refinance for the next house flip turned housing from a use commodity to a trading commodity. Arizona, California, Florida, and Nevada had the most pronounced housing cycles. Real estate investors accounted for nearly 45 percent of all home purchase lending at the peak of the boom in 2006, nearly double their share from 2000 [24]. In Las Vegas, Nevada, the speculative frenzy was particularly acute, with some condos trading multiple times before construction had even finished [24]. Gradually, the housing market became dependent on this continuous flipping. Although flippers were able to make easy money at first (2003-2005), this created the classic bubble pattern, where prices rose significantly beyond what local incomes could support because the market was overpowered by speculation—rather than people actually needing places to live.

2.4 Financial Innovation and Market Structure Changes

Prior to regulation changes, the traditional lending was as follows: the bank creates a loan, keeps the loan, and the bank also bears the loss if the borrower defaults, or is delinquent for more than 60 days in the first year. In the new Originate-to-Distribute (OTD) model, the bank would still create the loan,

but would now sell the loan/mortgage to third-party investment banks, such as Goldman Sachs, Lehman Brothers, etc. The investment banks would bundle hundreds and thousands of mortgages into mortgage-backed securities (MBS), which would then be resold to new investors worldwide, who could be insurance companies, pension funds, foreign banks, etc. [15, 3]

The policy was convenient for banks, who looked to minimize the amount of loans they offered on their balance sheets, allowing them to create more loans, therefore more profit. In the pursuit of increasing profit, banks stopped caring about who they were lending money to, as long as they could resell it to investment banks. People who otherwise wouldn't have been eligible for a loan through the traditional model did so through the new one, increasing the percentage of overall loan defaults [15, 3].

The Securities and Exchange Commission (SEC), the agency responsible for regulating the securities market and enforcing laws that prevent market manipulation, also became less aware, meaning that the agency allowed minimal documentation and criteria to be passed before the loans could be sold. However, this also meant that investment banks did not know how risky the mortgages they were packaging actually were. Due to the low regulations, banks could offer loans to people who were subprime and indirectly supply the Over-The-Counter (OTC) market [15, 3].

Similarly, credit rating agencies (Moody's, S&P, Fitch, etc.) can also be attributed to this lack of awareness with risky securities. Investors and those purchasing financial products rely heavily—to this day—on these ratings issued by Nationally Recognized Statistical Rating Organizations (NRSROs) to assess the quality of their investments and protect themselves from a possible downgrade in credit. From 2007-2008, 64% of all credit rating downgrades were attributed to securities that had home equity or first mortgages as collateral, with a large share belonging to ABS CDOs [6]. This type of CDO accounted for 42% global decreases in asset value reported by financial institutions [6]. As of October 2008, Citigroup, AIG, and Merrill Lynch took write-downs totaling \$34.1 billion, \$33.2 billion, and \$26.1 billion, respectively, due to exposure to ABS CDOs [6].

3 Policy-Driven Causes of Market Instability

3.1 The Federal Reserve

Preceding the collapse of the bubble, the United States experienced a period of low interest rates in the early 2000s. Following the dot-com market crash and the September 11 attacks, the Federal Reserve aggressively lowered the federal funds rate from over 6% in 2000 to nearly 1% by 2003, marking a historically low borrowing environment [22]. The rationale behind this move was to stimulate market activity through borrowing, spending, and investment. However, the

extended period of interest rates inadvertently encouraged expansion of credit and risk-taking, through both aggressive borrowing by consumers and lending by financial institutions.

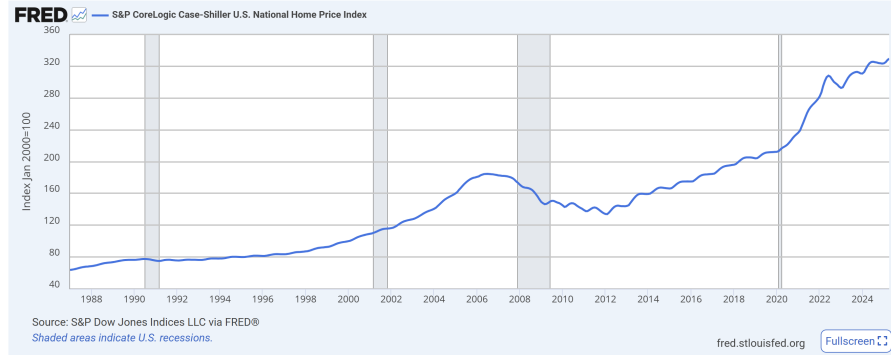


Figure 2: S&P/Case-Shiller U.S. National Home Price Index [9]

The surge in credit availability fueled rapid home price appreciation. As shown in Figure 2, the national home price index rose sharply from 2000–2006, significantly outpacing historical trends and suggesting speculative dynamics in the housing market [9].

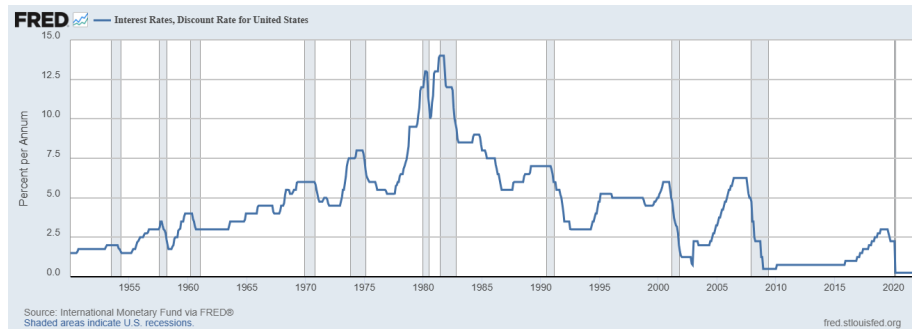


Figure 3: Federal Interest Rates (United States)

The Federal Reserve began to raise interest rates in 2004, reaching approximately 5.25% by mid-2006 [22]. The sudden increase in interest rates, as depicted in Figure 3, placed significant financial strain on households with ARMs, triggering payment shocks and a cascade of defaults. As the number of defaults rose, the value of MBSs dropped, revealing the inherent fragility of a financial system built on unsustainable lending [15, 3].

F.218 Home Mortgages (1)

Billions of dollars; quarterly figures are seasonally adjusted annual rates

	2004	2005	2006	2007	2008	2009	2009				2010	
							Q1	Q2	Q3	Q4	Q1	
1 Net borrowing	1013.5	1114.2	1073.2	710.3	-103.7	-211.9	-50.3	-222.5	-442.1	-132.6	-452.4	1
2 Household sector	926.0	1039.0	990.7	673.5	-46.6	-162.6	-8.7	-168.5	-359.6	-113.5	-389.8	2
3 Nonfinancial corporate business	4.7	7.6	8.2	2.9	-10.2	-12.4	-10.8	-13.0	-13.8	-12.0	-7.8	3
4 Nonfarm noncorporate business	82.8	67.6	74.2	33.9	-46.9	-36.9	-30.8	-41.1	-68.6	-7.1	-54.8	4
5 Net change in assets	1013.5	1114.2	1073.2	710.3	-103.7	-211.9	-50.3	-222.5	-442.1	-132.6	-452.4	5
6 Household sector	6.6	5.0	-15.0	-12.1	0.4	-8.0	-8.0	-8.0	-8.0	-8.0	-8.0	6
7 Nonfinancial corporate business	13.8	1.0	-5.1	-10.9	-4.4	-4.3	-4.3	-4.3	-4.2	-4.3	-4.2	7
8 Nonfarm noncorporate business	1.6	1.9	-0.6	2.7	0.9	-1.1	-0.9	-1.1	-1.4	-1.2	-1.2	8
9 State and local governments	4.2	5.5	7.4	5.3	-2.7	10.0	-7.6	20.1	22.3	5.3	4.6	9
10 Federal government	-0.5	-0.4	0.2	0.3	2.7	5.7	-0.2	10.2	15.6	-2.9	4.6	10
11 Commercial banking	221.9	210.5	290.1	128.4	30.0	10.7	4.7	135.0	-287.9	191.0	-285.2	11
12 Savings institutions	171.4	79.6	-86.0	11.1	-212.6	-217.7	-53.6	-416.9	-185.5	-214.9	-19.6	12
13 Credit unions	28.7	31.3	30.1	31.8	33.2	3.2	22.9	-4.5	-9.7	4.1	22.0	13
14 Life insurance companies	-0.4	-0.6	2.9	-0.9	0.8	-4.6	-1.9	-1.8	-1.6	-13.2	-0.2	14
15 Private pension funds	-0.2	-0.0	-0.1	-0.1	0.1	0.7	1.6	0.3	0.4	0.5	0.1	15
16 State and local govt. retirement funds	-1.9	0.5	-0.7	-1.7	-0.1	-0.0	-0.1	0.1	-0.1	-0.1	-0.0	16
17 Government-sponsored enterprises	-10.5	-54.7	3.7	-9.7	9.1	-18.4	50.8	-72.8	-34.6	-16.9	22.1	17
18 Agency- and GSE-backed mortgage pools	43.8	169.0	302.8	622.6	492.2	402.5	298.6	530.4	469.5	311.5	67.1	18
19 ABS issuers	382.2	573.3	518.9	33.4	-313.4	-333.7	-333.1	-386.2	-342.4	-273.2	-221.8	19
20 Finance companies	101.8	67.8	48.3	-65.4	-97.3	-47.8	-7.3	-19.6	-69.4	-94.8	-39.0	20
21 REITs	50.9	24.6	-23.7	-24.5	-42.5	-9.1	-12.1	-3.5	-5.2	-15.7	6.4	21
Memo:												
22 Home equity loans included above (2)	182.2	139.2	151.3	64.7	-18.0	-81.9	-70.2	-87.4	-93.1	-77.1	-110.8	22
23 Commercial banking	117.6	65.5	104.6	38.6	82.6	-14.3	-25.5	46.2	-57.9	-20.0	-74.8	23
24 Savings institutions	25.6	30.4	-14.1	42.9	-61.0	-39.5	-14.0	-103.8	-8.4	-31.8	-6.2	24
25 Credit unions	12.2	12.0	11.0	7.2	4.6	-4.1	-6.7	-3.7	-4.5	-1.5	-10.2	25
26 ABS issuers	7.1	17.1	40.1	-11.0	-24.6	-14.5	-18.3	-16.2	-13.6	-9.7	-11.0	26
27 Finance companies	19.7	14.3	9.7	-13.1	-19.5	-9.6	-5.6	-9.9	-8.7	-14.0	-8.5	27

(1) Mortgages on 1-4 family properties including mortgages on farm houses.

(2) Loans made under home equity lines of credit and home equity loans secured by junior liens. Excludes home equity loans held by individuals.

Figure 4: Home Mortgages from 2004–2010 (Q1) [1]

The effect of fluctuations in interest rates can be seen through an analysis of the net borrowing and asset changes related to home mortgages. In Figure 4, as the bubble expanded (2004–2006), Net Borrowing (Row 1) soared to \$1114.2B in 2005, a reflection of the housing boom and rapid mortgage expansion. Shortly after, Net Borrowing sharply *declined* by \$403.9B in 2007, showing the beginning of the credit crunch. Following the pattern of decline, Borrowing turns negative (-\$103.7B), and dips even lower in 2009 to -\$211.9B. The rapid shift from borrowing to deleveraging after 2006 further illustrates the effect of mass defaults and tightened lending standards [1].

3.2 Deregulation and Shadow banking

Although the formation of the bubble can be attributed to consumer behavior triggered by interest rates, there were also policies that allowed for much of this behavior to go unchecked.

In 1999, the Gramm–Leach–Bliley Act (GLBA) further blurred the line separating commercial and investment banking by repealing key provisions of the Glass–Steagall Act of 1933, which had originally been enacted in response to the Great Depression to limit speculative financial activity. The GLBA allowed commercial banks, investment banks, and insurance companies to consolidate

and engage in each other’s activities, contributing to the emergence of large and highly interconnected financial conglomerates [3, 13]. As these institutions expanded into riskier financial activities, many operated under the assumption that systemically important firms would likely receive government support during periods of severe financial distress, illustrating the broader problem of moral hazard within the financial system [15].

Another major deregulatory change in the banking system was the Riegle–Neal Interstate Banking and Branching Efficiency Act of 1994, which removed many restrictions on interstate banking and branching, allowing banks to rapidly expand across state lines [13]. The legislation accelerated consolidation within the financial sector, enabling institutions such as Bank of America, JPMorgan Chase, and Wells Fargo to expand aggressively into national markets [13]. As banking became increasingly concentrated, large financial institutions played a growing role in mortgage origination, securitization, and interconnected financial activities. This consolidation contributed to a system in which financial distress at a small number of major institutions could spread rapidly throughout the broader economy, amplifying the severity of the 2008 financial crisis [15, 3].

Apart from federal legislation, significant deregulatory measures were also implemented by the Office of the Comptroller of the Currency (OCC) and the Office of Thrift Supervision (OTS), both of which were responsible for supervising financial institutions in the United States. On January 13, 2004, the OCC issued the “Preemption and Visitorial Powers” rules, which expanded federal preemption authority over nationally chartered banks [4]. The Preemption Rule established that state laws could not apply to national banks if they “obstruct, impair, or condition” a bank’s federally authorized lending powers. As a result, many state anti-predatory lending laws were effectively prevented from applying to nationally chartered institutions, contributing to regulatory gaps that enabled increasingly risky mortgage lending practices [2, 3]. The Visitorial Powers Rule further restricted the ability of state authorities to inspect, supervise, regulate, or enforce compliance against national banks except where explicitly permitted under federal law, reinforcing the dominance of federal oversight during the housing boom [4, 15].

Similarly, in 2003, the OCC ruled that key provisions of the Georgia Fair Lending Act would be preempted for nationally chartered banks, despite the law’s intent to protect consumers from predatory lending practices through restrictions on loan terms, interest rates, and fees [4, 3]. This decision contributed to the emergence of a dual regulatory system in which nationally chartered banks operated under more lenient federal standards rather than stricter state consumer-protection laws. In a similar manner, the Office of Thrift Supervision (OTS) issued comparable preemption determinations for federally chartered thrifts, allowing them to bypass portions of Georgia’s lending regulations [3]. Comparable conflicts also emerged in states such as New York, where state au-

thorities attempted to enforce stronger anti-predatory lending standards while federal regulators asserted that federally supervised institutions were exempt from these state-level regulations [15, 3].

The OCC and OTS preemption of state anti-predatory lending laws in the early 2000s, including portions of Georgia’s Fair Lending Act, contributed to a regulatory environment that weakened state-level consumer protections during the housing boom [4, 3]. Although federal regulators argued that national oversight would provide consistent supervision of financial institutions, critics later contended that the federal approach relied heavily on a deregulatory “light-touch” philosophy that failed to adequately restrain risky lending practices [15, 13]. Compared with many state anti-predatory lending laws, federal standards were often less restrictive, and enforcement mechanisms proved insufficient as mortgage lending expanded rapidly. Scholars and post-crisis investigations also pointed to problems of regulatory capture, inadequate data collection, and failures to recognize growing systemic risks within mortgage and securitization markets [15, 6]. In retrospect, these preemption policies weakened the ability of states to enforce stronger consumer-protection standards during a critical period leading up to the 2008 financial crisis.

3.3 Oversight of Derivatives and Securitization

With the intention of promoting financial innovation and market efficiency, the Commodity Futures Modernization Act (CFMA) of 2000 clarified that many over-the-counter (OTC) derivatives would remain largely exempt from federal regulation by the Commodity Futures Trading Commission (CFTC) and outside significant oversight by the Securities and Exchange Commission (SEC) [3]. As a result, credit default swaps and other OTC derivative products expanded rapidly in largely opaque markets with limited transparency, reporting requirements, or centralized supervision [15]. In the years preceding the financial crisis, major institutions such as AIG and Lehman Brothers accumulated massive derivative exposures tied to mortgage-backed securities and structured credit products, often with insufficient capital buffers relative to the risks they assumed [15, 6]. Critics later argued that the limited regulation and transparency of OTC derivatives markets amplified systemic risk and contributed significantly to the severity of the 2008 financial crisis [15, 3, 6].

Yet the effort to stimulate financial markets did not stop there. In 2004, the SEC established the Consolidated Supervised Entities (CSE) program, partly to satisfy European Union requirements that major U.S. investment banks be subject to consolidated supervision under U.S. law [7, 15]. Under the program, the five largest investment banks — Bear Stearns, Goldman Sachs, Lehman Brothers, Merrill Lynch, and Morgan Stanley — were permitted to use internal risk models to determine regulatory capital requirements, enabling significantly higher leverage ratios than under prior standards [15]. From 2004 to 2007, these firms dramatically increased their leverage and exposure to mortgage-

related securities, leaving them highly vulnerable to declines in asset values and disruptions in short-term funding markets. Despite the SEC’s oversight framework, the CSE program failed to restrain excessive risk-taking, and by 2007 the five firms collectively held more than \$4 trillion in debt obligations [15, 7]. Ironically, the program intended to improve supervision ultimately coincided with the expansion of the very risks that contributed to the 2008 financial crisis.

4 The Crisis Unfolds (2007-2008)

4.1 Early signs

In 2007, subprime defaults and stress from financial institutions were all early signs of the financial crisis. The delinquency rate on subprime mortgages quickly rose from the 2% average to more than 7% by the last quarter of 2008 [8]. The ABX index, launched in 2006, was designed to provide information about the 20 largest mortgage-backed securities [11]. The ratings are categorized by credit quality levels (AAA to BBB-), with higher ratings indicating a healthy market. The AAA index, intended to be the most secure and valuable quality ratings, dropped by more than 30% in value from June 2007 to June 2008, with the BBB-minus dropping more than 90%, by June 2008 [11, 6].

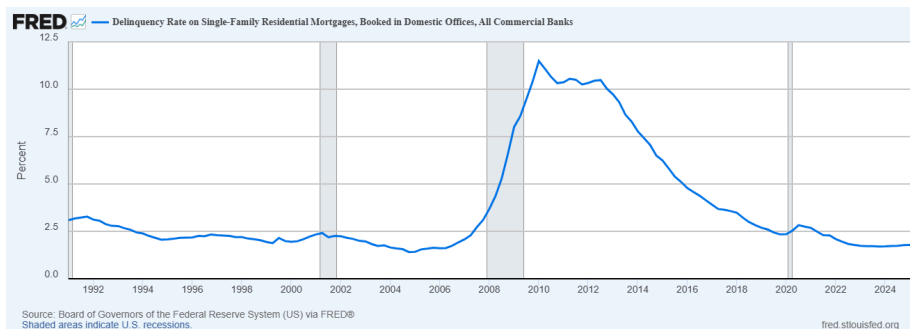


Figure 5: Delinquency Rate on Single-Family Residential Mortgages in the United States[8]

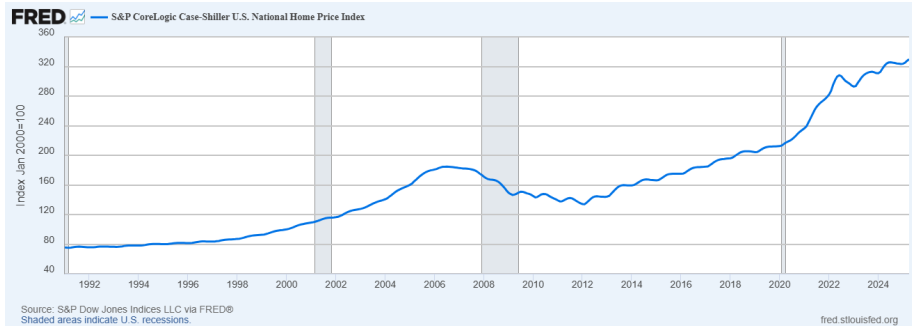


Figure 6: S&P/Case-Shiller U.S. National Home Price Index [9]

As shown in Figure 5 above, the rate of residential mortgage defaults quickly rose starting in 2006 [8]. In turn, the loan defaults suddenly lowered the demand for homes, with prices reaching a critical turning point when the national Case-Shiller index attained its all-time high of 198.01 in 2006 Q1, followed by a *significant* decline [9]. Across the country, home values typically fell 20% to 60%, peak to bottom, depending on the area and how badly it was affected by foreclosures [17, 15].

New Century, once one of the largest subprime lenders in the nation, ignored early warning signs that their loans were deteriorating, with 17% of loans going into default within 3 months of purchase. The issue originated from the company's risk-control department. From 2003-2005, there had been extreme underwriting errors, including predatory lending, legal and state violations, and credit issues in over 25% of the loans they had offered. As a result, investors demanded that their deteriorated loans be bought back by New Century. The financial pressure became overwhelming when New Century announced that if it had to buy back all of its mortgages, it could be on the hook for nearly \$8.5bn, and all its Wall Street lenders had abruptly cut off funding. The February 7, 2007 announcement stunned the market, wiping out over \$600 million in market value in a one-day 36% drop from \$30.16 to \$19.24 per share. This dramatic collapse of one of the nation's largest subprime lenders served as a critical early warning, showing how quickly mortgage lenders with deteriorating loan portfolios could face complete financial ruin [25].

4.2 Bank Failures and Liquidity Freeze

After commercial and investment banks merged through the GLBA Act of 1999, financial institutions also suffered. In the summer of 2007, two Bear Stearns hedge funds collapsed, as they were heavily invested in subprime MBS, signaling broader market issues [15, 13]. Shortly after, in September 2008, investment bank Lehman Brothers filed for bankruptcy, the largest in U.S. history with over \$639bn in assets [15, 13].

Another early signal of systemic distress emerged in the interbank lending markets, most notably through the widening of the LIBOR-OIS spread. The spread measures the difference between the London Interbank Offered Rate (LIBOR), the rate at which banks lend to one another, and the Overnight Indexed Swap (OIS) rate, which reflects the expected Federal Reserve policy rate and is considered near risk-free [12]. In normal market conditions, the spread averages around 10 basis points (0.10%), indicating healthy trust among banks. However, by August 2007, the spread began to spike sharply, surpassing 100 basis points by December, as fears of counter-party exposure to subprime mortgage-backed securities intensified [12, 15]. The surge in this spread revealed a sudden breakdown in interbank trust and a reluctance to lend, even overnight, marking a critical inflection point in the emerging credit crisis. This elevated spread persisted into 2008, becoming a key real-time indicator of deepening financial stress [15].

4.3 Stock Market Collapse and Global Shockwaves

The collapse of U.S. housing-related credit markets set off one of the most severe equity market declines in history. Investor panic began to accelerate after the September 15, 2008 bankruptcy of Lehman Brothers, which signaled that even systemically important institutions could fail [15, 13].

As reported by the Federal Reserve, from the October 2007 peak to March 9, 2009, the S&P 500 fell about 57%, signaling a broad loss of confidence in financial institutions and securitized credit markets [15, 26]. This drop in the index erased more than \$8 trillion in U.S. equity market capitalization, with the Dow Jones Industrial Average falling nearly 34% in just six months after Lehman’s collapse [15, 26]. The TED spread, a metric used to indicate credit risk in the U.S, spiked up to 4.64 percentage points on October 10, 2008, the normal existing at approximately 0.5 pp [10]. Yet this shock wasn’t limited to the nation. It spanned globally, with the World Trade Organization predicting a 9% drop in trade volume [27]. The MSCI World Index fell over 40% in 2008, and emerging market equity indices lost more than 50% in some cases [28]. Moreover, the global GDP decreased by an entire percentage point, as reported by IMF, further emphasizing the collateral damage of the crisis [28].

4.4 Emergency Policy Response and TARP

There were three main categories in which policymakers sought to issue immediate remedial measures in: financial system stabilization, economic stimulus, and regulatory reform.

- **Financial System Stabilization:** The first step was to stop the collapse of the banking sector and unfreeze the credit markets. Under the Troubled Asset Relief Program (TARP), the U.S. Treasury provided \$443.5 billion in capital and guarantees to banks, insurers, and auto companies, preventing further large-scale insolvencies [15, 13, 32]. As most banks struggled

to remain afloat, the government provided liquidity support in the form of lending facilities such as the Term Auction Facility (TAF), Commercial Paper Funding Facility (CPFF), and Primary Dealer Credit Facility (PDCF) to ensure banks and corporations could roll over short-term debt [15, 1].

To further instill confidence in the banks, FDIC deposit insurance limits were temporarily raised from \$100,000 to \$250,000 per depositor, and explicit guarantees were extended to certain types of bank debt to prevent runs [13, 15]. Lastly, the most important step was to eliminate the defaulted MBS from the credit markets. Also aimed at lowering mortgage rates and supporting housing finance, the Fed purchased \$1.25 trillion in mortgage-backed securities (MBS) between 2009–2010 [15, 34].

- **Economic Stimulus:** Although the previous measures stabilized the banks, what about the economy? The American Recovery and Reinvestment Act (ARRA), passed in February 2009, injected hundreds of billions of dollars into the economy via infrastructure projects, aid to states, tax cuts, and unemployment benefit extensions [15]. In order to stimulate lending and market activity once again, the Federal Reserve slashed the federal funds rate to 0–0.25% by December 2008 and committed to keeping rates near zero for an extended period [22, 34].

Yet it was equally important to gain confidence from investors. As a result, large-scale purchases of Treasuries and agency securities (QE1, QE2, QE3) were aimed to lower long-term interest rates, stimulate borrowing, and encourage investment [15, 34]. To further target the housing market, programs like the Home Affordable Modification Program (HAMP) and Home Affordable Refinance Program (HARP) were designed to prevent foreclosures and stabilize housing markets [15, 13].

- **Regulatory Reform:** Once these immediate solutions were put into action, the government sought to create a system that would prevent a similar crisis from happening again. One such measure was the Dodd–Frank Wall Street Reform and Consumer Protection Act (2010). Another was establishing the Consumer Financial Protection Bureau (CFPB) to regulate mortgage lending and consumer finance products. The U.S. government also created the Financial Stability Oversight Council (FSOC) to monitor systemic risks.

Lastly, to tackle the lack of transparency, there were now higher capital and liquidity standards on large banks and securitizers, mandating central clearing for many derivatives [15, 13, 5]. In order to further identify if a financial institution is on the brink of collapse, the Supervisory Capital Assessment Program (SCAP) in 2009 was introduced, which would stress test for large financial institutions, later institutionalized as the Comprehensive Capital Analysis and Review (CCAR) [15, 13]. Though the issue wasn't just transparency, it was also speculative behavior. Written in

2010, the Volcker Rule restricted banks from engaging in proprietary trading and certain relationships with hedge funds and private equity funds to reduce speculative risk-taking [15, 13].

5 Consequences and Long-term Effects

5.1 Housing Market Collapse and Foreclosures

The bursting of the U.S. housing bubble triggered one of the steepest declines in home prices in modern history. After peaking in mid-2006, the S&P/Case-Shiller 20-City Composite Home Price Index fell about 33% by early 2012, wiping out trillions of dollars in household wealth [9, 15]. In some metropolitan areas—such as Las Vegas, Phoenix, and Miami—prices dropped by more than 50%, reflecting just how inflated values had become during the boom years [17, 15]. Nationally, home prices fell roughly 27–30% peak to trough [9, 15].

As prices declined, millions of homeowners slipped into negative equity, meaning their mortgage balance exceeded their home’s market value. By the fourth quarter of 2009, CoreLogic estimated that around 26% of mortgaged properties—some 11 to 12 million households—were underwater [15]. This created a dangerous feedback loop: homeowners unable to refinance or sell became more likely to default, and each foreclosure added supply to a market already glutted with distressed properties, pushing prices even lower [15, 14].

Foreclosures surged to unprecedented levels. RealtyTrac reported about 2.8 million properties with foreclosure filings in 2009 and 2.9 million in 2010 [29, 30]. States like Nevada, Arizona, Florida, and California were hardest hit, with Nevada reaching foreclosure rates as high as one in every nine homes in 2009 [29]. Beyond the economic damage, these waves of foreclosures devastated communities, hollowed out tax bases, and increased demand for local government services just as revenues were collapsing [15, 13].

5.2 Bank Failures

The collapse in housing prices and the implosion of mortgage-backed securities markets inflicted catastrophic losses on financial institutions across the spectrum. Large investment banks had built up massive exposures to subprime mortgages, collateralized debt obligations (CDOs), and other structured credit products that became nearly worthless as defaults rose [15, 6, 13].

The most visible failures became milestones of the crisis: Lehman Brothers filed for bankruptcy on September 15, 2008, the largest in U.S. history with \$639 billion in assets [15, 13]. Just ten days later, Washington Mutual was seized by regulators in the largest-ever U.S. commercial bank failure, with \$307 billion in assets [13, 15]. Bear Stearns was sold to JPMorgan Chase in March 2008 for

a fraction of its former value, and Merrill Lynch was forced into a merger with Bank of America [15, 13].

Failures weren't limited to Wall Street. From 2008 to 2012, the Federal Deposit Insurance Corporation (FDIC) closed 465 banks, peaking at 157 closures in 2010 [13]. Many of these were regional or community banks with concentrated exposure to construction loans and commercial real estate—both of which collapsed alongside the housing market [31, 13]. As counterparty risk fears spread, short-term interbank lending froze, and the Federal Reserve had to provide unprecedented liquidity support to keep basic market functions alive [15, 12].

5.3 Recovery Measures and Changes in Legislation

Policymakers responded with a combination of immediate stabilization measures and long-term reforms. The initial focus was to prevent a complete systemic collapse. The Emergency Economic Stabilization Act of 2008 authorized the \$700 billion Troubled Asset Relief Program (TARP), later reduced to \$475 billion under Dodd–Frank [32]. While originally designed to purchase toxic mortgage-related assets, TARP was used primarily for direct capital injections into banks, insurers, and even the auto industry. Total TARP disbursements reached about \$443.5 billion, with a net lifetime cost of roughly \$31.1 billion after repayments and investment income [32, 33].

The Federal Reserve complemented these actions with aggressive monetary policy. By December 2008, interest rates were slashed to 0–0.25%, and large-scale asset purchases—commonly known as Quantitative Easing—were launched to drive down long-term borrowing costs [22, 34]. Between late 2008 and 2010, the Fed purchased \$1.25 trillion in mortgage-backed securities, \$200 billion in agency debt, and \$300 billion in longer-term Treasuries [34]. FDIC deposit insurance limits were temporarily raised from \$100,000 to \$250,000, helping to restore depositor confidence, while emergency facilities stabilized commercial paper and repo markets [13, 15].

Long-term reforms addressed the systemic weaknesses exposed by the crisis. The Dodd–Frank Wall Street Reform and Consumer Protection Act of 2010 created the Consumer Financial Protection Bureau (CFPB) to oversee mortgage lending and consumer finance practices, established the Financial Stability Oversight Council (FSOC) to monitor systemic risks, and imposed higher capital and liquidity requirements on large banks [35]. It also mandated annual stress tests and “living wills” for the largest financial institutions, introduced the Volcker Rule to limit speculative proprietary trading, and reformed mortgage markets with stricter underwriting standards and greater transparency [35, 15]. These measures were designed to strengthen the resilience of the financial system and reduce the likelihood of another crisis of this scale.

6 Forecasting Financial Bubbles with Machine Learning

To better understand the key drivers of the 2008 Housing Bubble, we will be taking a closer look at Los Angeles, California, a populous city that experienced the sudden crash in full effect.

We will be analyzing these factors through an experimental and data-driven method, by forecasting the values of the S&P CoreLogic Case-Shiller CA-Los Angeles Home Price Index (LXXRSA). As previously defined, a housing bubble comprises of a rapid increase in home prices fueled by factors such as interest rates, consumer sentiment, geographic location, and more. For this reason, the Home Price Index is a good indicator of overall real estate market health, since it closely tracks this rise and fall of home prices through a standardized index.

The HPI metric is calculated by tracking prices of the *same* single-family homes over time through the Robust Interval and Value Weighted Arithmetic Repeat Sales (IVWARS) algorithm, or Repeated Sales Method. It records the prices of these single-family homes that are representative of the area and have been sold at least twice. The purpose of choosing these sale-pairs is to capture the price appreciation or depreciation of the same property. Then, the values are inputted into a weighted ordinary least squares (WLS) regression.

$$P_{i,t_2} - P_{i,t_1} = \sum_{k=1}^T \beta_k (D_{k,t_2} - D_{k,t_1}) + \varepsilon_i \quad (1)$$

In Figure 1, P_{i,t_1} and P_{i,t_2} are the sale prices of home i on different days. Respectively, D_{k,t_1} and D_{k,t_2} are the times at which the home has been sold. The regression coefficient, β_k , captures the month-by-month appreciation/depreciation given a baseline. By definition, the model operates in a way that gives less importance, or weight, to sale-pairs with long time gaps or extreme price changes. This type of model is known as a heteroscedastic weight scheme.

Once regression coefficient β_k is calculated, the index for month t is as follows:

$$Index_t = 100 \times \frac{\beta_t}{\beta_{Jan2000}} \quad (2)$$

Figure 2 represents how the final index value is calculated, assuming a baseline of January 2000. This baseline year is chosen reference a point to which we can compare price changes to, usually at a time when the housing market is neutral and stable. For example, we would not choose any month from the period of 2002-2009, as it encompassed both the housing boom and crash.

7 Data Sources and Feature Engineering

In this prediction model, we have selected fifteen input variables obtained from FRED (Federal Reserve Economic Data from Federal Reserve Bank of St. Louis) and CAR (California Association of Realtors). Our choice for the following variables revolves around avoiding both collinearity and association with the target variable, indicated in italics.

- 30-year Fixed Rate Mortgage Average in United States (FRED)
- All Employees Retail Trade in California (FRED)
- Consumer Price Index for All Urban Customers in U.S. City Average (FRED)
- Federal Funds Effective Rate (FRED)
- Household Debt Service Payments as a Percent of Disposable Income (FRED)
- Labor Force Participation Rate for California (FRED)
- Market Yield on U.S. Treasury Securities at 10-Year Constant Maturity (FRED)
- Median Prices of Existing Detached Homes Historical Data (CAR)
- Median Time on Market Existing Detached Homes Historical Data (CAR)
- New Private Housing Units Authorized by Building Permits for California (FRED)
- New Privately-Owned Housing Units Started Total Units (FRED)
- Percent Change of Sales of Existing Detached Homes Historical Data (CAR)
- Real Gross Domestic Product (FRED)
- *S&P CoreLogic Case-Shiller CA-Los Angeles Home Price Index (FRED)*
- Unemployment Rate in California (FRED)
- University of Michigan Consumer Sentiment (FRED)
- Unsold Inventory Index of Existing Detached Homes Historical Data (CAR)

These variables were selected in a manner to consider the demand-side indicators, supply-side indicators, housing market-specific conditions and macroeconomic controls as discussed previously.

To prepare the data for the model, we must consider both the frequency and range of each variable. All of the variables listed above were reported at a monthly frequency besides Household Debt Services as a Percent of Disposable Personal Income and Real Gross GDP, both reported every quarter, and 30-year Fixed Rate Mortgage Average in United States, reported every week.

To convert the quarterly frequency datasets into monthly, we applied a method of linear interpolation, which estimates the value between two known values given the assumption that the change between them is linear. A similar approach was used to resample the weekly data to a monthly frequency by aligning it to the first day of each month and interpolating any missing values. Upon the completion of this calculation and further analysis, we determined that all of the datasets overlap from roughly 02-01-1990 to 04-01-2025. The pandas package in Python was essential for both specifying the range for the time series and interpolating the desired data points.

7.1 Model Design and Evaluation

After consolidating the individual files into a dataset using the tidyverse and lubridate packages in R, we constructed a Long Short-Term Memory (LSTM) model in Python. The choice of this neural network, as opposed to regression and tree-based models, was dictated by how the model reacts to patterns over time [16]. As an LSTM model is designed to capture long-term *temporal* dependencies in sequential data (like monthly HPI values), it has the ability to model the seasonality, trend shifts and autocorrelation ever-so present in target variable. With many of the future values contingent on past ones, the LSTM model’s internal memory captured this delayed cause-effect relationships better than the other two choices.

To begin, we imported the libraries that would help us load and preprocess the data: pandas, numpy, matplotlib, keras, and sklearn. Prior to constructing the model, we must adjust the target variable. By definition, the LSTM model performs best when it considers stationary data. In other words, the Home Price Index has a long-term *non-stationary* upward trend, which can make it difficult to detect short-term fluctuations and rapid changes. To address this issue, we calculated the difference between consecutive target values, instead of the raw target, allowing the model to focus on these smaller and more directional shifts.

Yet, another issue is present in this new target column and its corresponding features: their scaling. When variables have different ranges, those with larger magnitudes can dominate the learning process, biasing the model and obscuring the true influence of other features. To solve this issue as well, we first split the data and then reduced this likelihood using the MinMaxScaler() function, which converts each data point to a value in the range (0,1). It is very important to split the data *first* and then scale it, to prevent data leakage.

Now, using the keras package, we construct the sequential LSTM model. We included specific layers to better suit our purpose: to detect both long-term and short-term trends in the HPI data. To help our model learn from both past and future context, we included a bidirectional layer, which by definition processes sequences in forward and backward directions. But this influx of information may cause overfitting. To account for this issue, we included multiple dropout layers evenly spaced throughout the model’s architecture to prevent it from memorizing the training data points too closely. Lastly, we included dense layers, which takes all outputs from the previous layer and applies a weighted sum. The primary purpose behind including this layer is to help the model learn more complex and non-linear relationships.

To compile the model, we utilized an Adam optimizer, which adjusts the learning rate (how quickly the model updates its weights) during the training period and allows a more stable convergence to the final value. As far as the loss method, we chose the Huber loss method. As housing prices can tend to have sudden spikes and outliers, Huber Loss is a mix of MSE and MAE, balancing both precision and sensitivity to outliers.

Throughout the creation of this LSTM. model, we controlled for the following variables:

- **Seed:** maintains reproducibility of results by setting a random seed that ensures consistent model behavior
- **Lookback:** how much history (in months) the model sees at each step (the longer the lookback, the more temporal context)
- **Test Size:** amount of data reserved for evaluating the model
- **Batch Size:** number of samples processed in each training batch, with a smaller batch size allowing the model to update the weights more frequently
- **Epochs:** number of training cycles
- **Dropout:** fraction of neurons dropped randomly, to prevent overfitting
- **Patience:** number of epochs that must pass with no model improvement before early stopping

7.2 Results and Interpretation

After training our model and predicting on the holdout set, we evaluated our results using R^2 , MAE, and RMSE metrics. Our results were as follows:

$$R^2 = 0.9770, \quad \text{MAE} = 7.2284, \quad \text{RMSE} = 9.0807$$

$$R^2 = 1 - \frac{RSS}{TSS} \quad (3)$$

As shown in Equation 3, the R^2 indicates how well the model explains the variability of the test data, where RSS is the sum of the residuals squared and TSS is the total sum of squares. Typically, this value ranges from 0 to 1, a value of 1 indicating an excellent fit. In some cases, the value returned may be *negative*, which means that the model did worse than if it were to pick the average of each data point.

$$MAE = \frac{\sum_{i=1}^n |y_i - x_i|}{n} \quad (4)$$

The mean absolute error is calculated by taking the absolute value of the differences in predicted and true values. As far as its range, it is always above 0 and it is an unweighted metric, meaning that it is *not* sensitive to outliers.

$$RMSE = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{N - P}} \quad (5)$$

The last metric we used to evaluate the accuracy of the model was the RMSE. This takes root of the mean squared error of the model, yet unlike the last metric, it *is* sensitive to large values. Both this metric and the previous one are measured in the same units and follow the same convention: the lower the MAE and RMSE, the less mistakes the model is making.

However, although the LSTM model predicts the HPI index well, what does this tell us about the 2008 Housing Bubble Crisis? As previously mentioned, the LSTM model assigns specific weights to all the variables, with a higher weightage indicating that the model is more sensitive to that particular variable. To view these details, we applied a gradient-based feature importance method to the model.

$$\frac{\partial \hat{y}}{\partial x_{i,t}} \quad (6)$$

Using keras' TensorFlow package, we calculated partial derivatives, or gradients, of the model's output with respect to each of the fifteen features. By definition, a gradient asks the question, "How much will the function change if I nudge this one variable?" Gradients, as shown in Equation 6, represent the how sensitive the model is to each feature. By averaging the gradients over all time steps and samples, we quantified each feature's influence on the model's accuracy. Lastly, after normalizing these values and plotting them, we were able to view how much a *tiny* change in each feature affected the model.

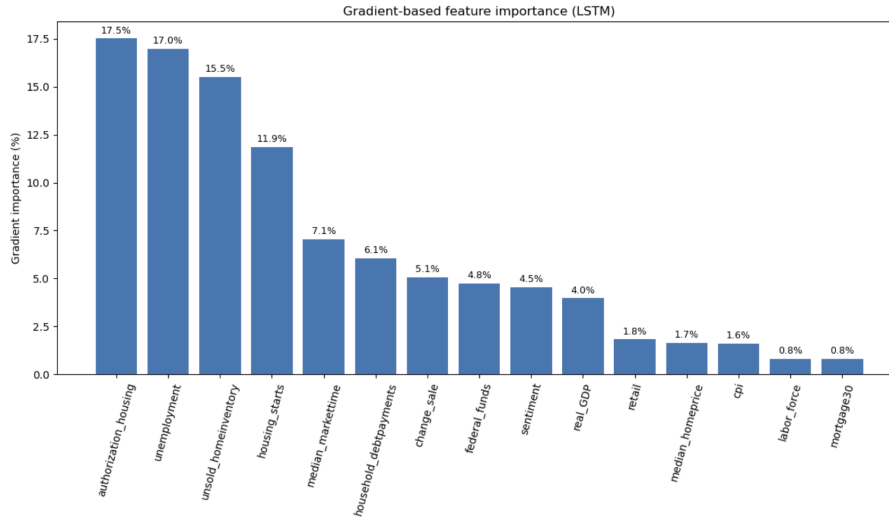


Figure 7: LSTM Model Feature Importance

In Figure 7 above, the bars represent the relative contribution of each input feature to the model’s predictions, expressed as a percentage of total influence. Features like authorization of housing (17.5%), unemployment (17.0%), and unused home inventory (15.5%) emerged as the most impactful, indicating that the model was highly sensitive to changes in these variables. In contrast, features such as 30-year Fixed Rate Mortgage (mortgage30), labor force, and Consumer Price Index (CPI) contributed less than 1%, suggesting they played a minimal role in shaping the model’s output. This analysis provides insight into which variables the LSTM model relied on most when forecasting housing market trends.

7.3 Risks and Limitations of ML Forecasting

Using machine learning to make predictions offers powerful tools for recognizing patterns in large datasets, but it also comes with important risks and limitations. One major concern is overfitting, where the model learns patterns that exist only in the training data rather than generalizable trends. This can lead to excellent performance on past data but poor results on new or unseen economic situations.

Another key limitation is data leakage, which occurs when information from the test set or future data accidentally influences the training process. This creates an illusion of high model accuracy, but it breaks the real-world assumption that future outcomes are unknown during prediction. This makes it incredibly important to be careful of how one splits, trains and scales their data.

Machine learning models, especially complex ones like LSTMs, also pose questions in terms of interpretability. Unlike traditional statistical models, these models often operate as "black boxes," making it difficult to understand or explain why a certain prediction was made. This limits their usefulness in high-stakes domains where decisions must be justified. Finally, ML models rely entirely on historical data and struggle with unseen events or structural breaks, such as financial crises, pandemics, or sudden policy changes. These weaknesses highlight the importance of using ML as a tool and always combining it with both *contextual knowledge* and *caution*.

8 A Discussion of Current Events

To understand how these findings derived from the 2008 Financial Crisis can be applied to the world today, we must carefully analyze the current values of the fifteen features we included in the LSTM model along with the ever-changing economic context to them.

8.1 Today's indicators

- **Labor Market Indicators:** In the run-up to the 2008 housing bubble, employment in retail trade and other consumer-facing sectors expanded steadily, supported by rising home equity and abundant credit. When the bubble burst, unemployment surged into double digits, and retail jobs in California fell by over 200,000. COVID, however, created a very different labor shock. In April 2020, California retail employment collapsed by nearly half a million jobs in a single month. It was a sharper fall than during 2008, but the recovery was faster, fueled by stimulus checks and pent-up demand. Today, unemployment is historically low compared to the Great Recession, but the labor force has not fully rebounded from the COVID time period, reflecting lingering effects of COVID-era workforce exits and demographic shifts [36, 37].
- **Consumer Prices and Inflation:** The 2008 housing bubble was not an inflationary event. Consumer prices grew moderately, and the collapse of demand in 2008–09 even created a brief deflationary scare. By contrast, COVID's aftermath unleashed the sharpest burst of inflation since the 1970s. Supply chain disruptions, fiscal stimulus, and shifts in consumer demand pushed the CPI up by more than 20 percent from 2020 to 2024. Unlike 2008, where the Federal Reserve cut rates to zero to fight deflation, today's policymakers have raised rates aggressively to combat inflation, creating a very different set of risks [38, 22].
- **Debt and Interest Rates:** Household debt service burdens peaked in 2007, with mortgages consuming record shares of disposable income. When housing values collapsed, households were left with underwater mortgages, triggering mass defaults. During COVID, household balance

sheets were supported by relief programs, low interest rates, and temporary forbearance. As a result, debt burdens fell to multi-decade lows by 2021. However, the Fed’s rapid rate hikes since 2022 have reversed that trend, making mortgages and credit cards more expensive. Unlike 2008, today’s stress is less about delinquency and more about affordability: new buyers are priced out as mortgage rates sit above 7 percent [39, 22, 20].

- **Housing Market Conditions:** The housing sector was ground zero in 2008, with falling Case-Shiller indices, rising foreclosures, and excess inventory driving the downturn. Today, the story is different. During COVID, record-low mortgage rates ignited a home-buying frenzy, sending California home prices to all-time highs. That momentum has since slowed sharply. Home prices remain elevated, but transactions are down, time on market has increased, and new housing starts have softened. The constraint today is supply, not oversupply: homeowners with 3 percent mortgages are reluctant to sell, leaving buyers to face tight inventory at much higher borrowing costs [9, 20].
- **Consumer Confidence and GDP:** Consumer sentiment plunged in 2008 as wealth evaporated and unemployment surged, reinforcing the recession. COVID brought an even sharper collapse in confidence in spring 2020, though unprecedented fiscal and monetary support prevented a prolonged contraction. Today, GDP growth is positive but uneven, and sentiment remains fragile. Consumers are squeezed between inflation, higher debt servicing costs, and the memory of pandemic instability [40, 41, 38].

8.2 Commercial Real Estate Risks

Commercial Real Estate (CRE) represents one of the most pressing financial vulnerabilities today. During the 2008 housing crisis, the epicenter of risk was residential housing, where collapsing prices, excess supply, and over-leveraged borrowers triggered systemic collapse [15, 13]. By contrast, today’s CRE market faces stress from a different set of pressures.

The COVID-19 pandemic permanently altered demand for office and retail space. Remote and hybrid work have reduced occupancy rates in urban office districts, while e-commerce has eroded brick-and-mortar retail demand. As leases expire, many landlords face refinancing challenges at much higher interest rates, raising the risk of defaults [22, 20]. Banks, particularly regional lenders with high exposure to commercial property loans, are vulnerable to a downturn in CRE valuations [31]. While not yet systemic, the parallels to 2008 lie in the potential for a localized asset shock to ripple through financial institutions and credit markets [15].

8.3 Tech Valuations and AI-Led Speculation

Another area of risk is concentrated in financial markets, particularly in technology stocks and the hype surrounding artificial intelligence. During the early

2000s, the dot-com bust showed how speculative valuations, disconnected from underlying earnings, could lead to a sudden collapse in investor wealth [28]. In 2008, overconfidence in housing markets and mortgage-backed securities played a similar role [15, 6].

Today, the rapid rise of AI-driven companies has created a new form of speculative concentration. While many firms are genuinely innovating, valuations of leading tech giants have expanded dramatically, often priced on expectations of future dominance rather than present profitability [40, 41]. This speculative dynamic carries echoes of past bubbles: excess liquidity, investor herding, and technological optimism fueling asset price inflation [15]. Should interest rates remain high or AI adoption prove slower than expected, a sharp repricing of tech equities could amplify volatility across broader financial markets, much like mortgage securities did during the housing bust [22, 28].

8.4 Credit Cycles, Interest Rates and Liquidity

The 2008 housing bubble was ultimately a story of credit excess. Ultra-loose lending standards, securitization of risky mortgages, and a global glut of liquidity combined to fuel unsustainable growth in housing. When the cycle turned, liquidity evaporated, and credit froze.

Today, credit dynamics look very different but carry their own risks. Instead of lax credit, households and firms face tight financial conditions due to the Federal Reserve’s aggressive rate hikes since 2022. The federal funds rate has climbed to levels not seen in two decades, and 10-year Treasury yields have surged [22]. This has sharply increased the cost of mortgages, corporate borrowing, and consumer credit. Liquidity conditions are tighter, yet global financial markets remain flush with capital seeking high returns, creating pockets of speculative excess.

The key systemic question mirrors 2008: can the financial system absorb these shifts without triggering cascading defaults? In 2008, the combination of rising defaults and evaporating liquidity broke the system. Today, the danger lies in whether elevated interest rates expose hidden vulnerabilities — in corporate debt, commercial real estate, or highly leveraged financial institutions — that could similarly destabilize credit markets.

9 Conclusion

The 2008 housing bubble exposed how excess credit, lax regulation, and misplaced confidence in rising home values could result in a systemic collapse. Its deflationary shock (mass foreclosures, unemployment, and a freeze in liquidity) reshaped the concept financial regulation and left a lasting mark on global markets. Comparably, COVID-19, the more recent globally catastrophic event,

created a very different disruption: a sudden halt in activity followed by aggressive stimulus, which gave rise to the highest inflation in decades and a new cycle of affordability pressures.

Today's indicators show that while the housing market no longer sits at the center of risk in the same way it did in 2008, vulnerabilities still exist. Retail employment and consumer sentiment reveal lingering fragility, inflation and high interest rates have eroded affordability, and structural shifts have left commercial real estate and credit markets under pressure. At the same time, speculative forces in technology mirror the excess optimism that fueled past bubbles.

The lesson from 2008 is that crises rarely repeat in identical form but often emerge from familiar dynamics: excessive leverage, overconfidence, and systemic blind spots. Whether through housing, pandemic aftershocks, or speculative surges in new technology sectors, financial stability remains contingent on awareness of these indicators and the ability to adapt policy before imbalances become systemic.

References

- [1] Federal Reserve Board. *Flow of Funds Accounts of the United States*. 2010. Available at: <https://www.federalreserve.gov/releases/z1/20100610/z1.pdf>
- [2] Federal Register. *Alternative Mortgage Transaction Parity Act Preemption*. 2002. Available at: <https://www.federalregister.gov/documents/2002/09/26/02-24407/alternative-mortgage-transaction-parity-act-preemption>
- [3] Cornell Law School. *Subprime Mortgage Regulation Analysis*. Available at: <https://scholarship.law.cornell.edu/cgi/viewcontent.cgi?article=1824&context=facpub>
- [4] Office of the Comptroller of the Currency. *OCC Bulletin 2004-6*. 2004. Available at: <https://www.occ.gov/static/rescinded-bulletins/bulletin-2004-6.pdf>
- [5] Bank for International Settlements. *History of the Basel Committee*. Available at: <https://www.bis.org/bcbs/history.htm>
- [6] National Bureau of Economic Research. *The Credit Rating Crisis*. 2010. Available at: <https://www.nber.org/reporter/2010number1/credit-rating-crisis>
- [7] U.S. Securities and Exchange Commission. *SEC Emergency Actions During Financial Crisis*. 2008. Available at: <https://www.sec.gov/news/press/2008/2008-230.htm>
- [8] FRED Economic Data. *Delinquency Rate on Single-Family Residential Mortgages*. Available at: <https://fred.stlouisfed.org/series/DRSFRMACBS>
- [9] FRED Economic Data. *S&P/Case-Shiller U.S. National Home Price Index*. Available at: <https://fred.stlouisfed.org/series/CSUSHPINSA>
- [10] FRED Economic Data. *TED Spread*. Available at: <https://fred.stlouisfed.org/series/TEDRATE>
- [11] Investopedia. *Behavior of the ABX Index During the 2008 Crisis*. Available at: <https://www.investopedia.com/ask/answers/120514/how-did-abx-index-behave-during-2008-subprime-mortgage-crisis.asp>
- [12] Bank for International Settlements. *The Global Financial Turmoil*. 2008. Available at: https://www.bis.org/publ/qtrpdf/r_qt0806u.htm
- [13] Federal Deposit Insurance Corporation. *History of the Financial Crisis*. Available at: <https://www.fdic.gov/bank/historical/crisis/chap1.pdf>

- [14] Mian, Atif and Sufi, Amir. *The Consequences of Mortgage Credit Expansion*. NBER Working Paper No. 15045. 2009. Available at: https://www.nber.org/system/files/working_papers/w15045/w15045.pdf
- [15] Financial Crisis Inquiry Commission. *The Financial Crisis Inquiry Report*. 2011. Available at: <https://www.govinfo.gov/content/pkg/GPO-FCIC/pdf/GPO-FCIC.pdf>
- [16] Fischer, Thomas and Krauss, Christopher. *Deep Learning with Long Short-Term Memory Networks for Financial Market Predictions*. 2017. Available at: <https://arxiv.org/abs/1709.08432>
- [17] Mayer, Christopher. *Housing Bubbles: A Survey*. Columbia Business School. Available at: <https://business.columbia.edu/sites/default/files-efs/pubfiles/5636/MayerHousingBubbles.pdf>
- [18] U.S. Census Bureau. *Historical Homeownership Rates*. Available at: <https://www.census.gov/housing/hvs/data/histtabs.html>
- [19] U.S. Department of Housing and Urban Development. *HUD Increases Housing Goals for Fannie Mae and Freddie Mac*. 2004. Available at: <https://archives.hud.gov/news/2004/pr04-038.cfm>
- [20] FRED Economic Data. *30-Year Fixed Rate Mortgage Average in the United States*. Federal Reserve Bank of St. Louis. Available at: <https://fred.stlouisfed.org/series/MORTGAGE30US>
- [21] Consumer Financial Protection Bureau. *The Mortgage Market and the Financial Crisis*. Available at: <https://www.consumerfinance.gov/>
- [22] FRED Economic Data. *Federal Funds Effective Rate*. Federal Reserve Bank of St. Louis. Available at: <https://fred.stlouisfed.org/series/FEDFUNDS>
- [23] Bernanke, Ben S. *The Global Saving Glut and the U.S. Current Account Deficit*. Federal Reserve Board, 2005. Available at: <https://www.federalreserve.gov/boarddocs/speeches/2005/200503102/>
- [24] Haughwout, Andrew F., Donghoon Lee, Joseph Tracy, and Wilbert van der Klaauw. *"Flip This House": Investor Speculation and the Housing Bubble*. Liberty Street Economics, Federal Reserve Bank of New York, December 5, 2011. Available at: <https://libertystreeteconomics.newyorkfed.org/2011/12/flip-this-house-investor-speculation-and-the-housing-bubble/>
- [25] Missal, Michael J. *Final Report of Michael J. Missal Bankruptcy Court Examiner*. United States Bankruptcy Court, 2008. Available at: https://fcic-static.law.stanford.edu/cdn_media/fcic-docs/2008-02-29%20Missal%20-%20Final%20Report%20on%20the%20Bankruptcy%20of%20New%20Century.pdf

- [26] Macrotrends. *Historical S&P 500 Data*. Available at: <https://www.macrotrends.net/>
- [27] World Trade Organization. *World Trade 2009 Prospects: Trade to Collapse by 9% in 2009*. 2009. Available at: https://www.wto.org/english/news_e/pres09_e/pr554_e.htm
- [28] International Monetary Fund. *World Economic Outlook: Crisis and Recovery*. 2009. Available at: <https://www.imf.org/external/pubs/ft/weo/2009/01/>
- [29] RealtyTrac. *Year-End Report Shows Record 2.8 Million U.S. Properties With Foreclosure Filings in 2009*. 2010. Available at: https://fcic-static.law.stanford.edu/cdn_media/fcic-docs/2010-01-14%20RealtyTrac%20Year-End%20Report%20Shows%20Record%202.8%20Million%20US%20Properties%20with%20Foreclosure%20Filing.pdf
- [30] RealtyTrac. *2010 Year-End U.S. Foreclosure Market Report*. 2011. Available at: <https://mytowncryer.com/wp-content/uploads/2011/02/realtytrac-year-end-2010-foreclosure-sales-data-final.pdf>
- [31] Mendes & Mount LLP. *The Impact of the Financial Crisis on Community Banks*. Available at: <https://mendes.com/news/the-impact-of-the-financial-crisis-on-community-banks/>
- [32] U.S. Department of the Treasury. *Troubled Asset Relief Program (TARP)*. Available at: <https://home.treasury.gov/data/troubled-asset-relief-program>
- [33] U.S. Government Accountability Office. *Troubled Asset Relief Program: Lifetime Cost*. 2023. Available at: <https://www.gao.gov/products/gao-24-107033>
- [34] Federal Reserve Board. *Federal Reserve Announces Further Actions to Support Credit Markets and the Economy*. 2008. Available at: <https://www.federalreserve.gov/newsevents/pressreleases/monetary20081125b.htm>
- [35] U.S. Congress. *Dodd-Frank Wall Street Reform and Consumer Protection Act*. 2010. Available at: <https://www.congress.gov/bill/111th-congress/senate-bill/3217>
- [36] U.S. Bureau of Labor Statistics. *Employment Situation Summary*. Available at: <https://www.bls.gov/>
- [37] Federal Reserve Economic Data (FRED). *Civilian Unemployment Rate and Labor Force Participation Rate*. Available at: <https://fred.stlouisfed.org/>

- [38] Federal Reserve Economic Data (FRED). *Consumer Price Index for All Urban Consumers (CPI-U)*. Available at: <https://fred.stlouisfed.org/>
- [39] Federal Reserve Economic Data (FRED). *Household Debt Service Payments as a Percent of Disposable Personal Income*. Available at: <https://fred.stlouisfed.org/>
- [40] Federal Reserve Economic Data (FRED). *University of Michigan Consumer Sentiment Index*. Available at: <https://fred.stlouisfed.org/>
- [41] Federal Reserve Economic Data (FRED). *Gross Domestic Product*. Available at: <https://fred.stlouisfed.org/>